



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 6: Consumer Theory IV

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ICS2523 Microeconomics

Lecture 6

Last lecture we

- Learnt the difference between **Marshallian and Hicksian Demand**
- Reviewed the duality in the consumer theory
- Reviewed some exercises

In today's lecture, we will

- Learn some important identities
- Analyse **changes in Income and Prices**
- Learn the **Slutsky equation**
- Do exercises to practice these new concepts

Announcement

Test 1 (Thursday, April 3rd)

- You may have a **cheat sheet** (formulario), but you will work on it collaboratively in a Canvas Forum.
- As you study, you may post there the relevant formulas (not solutions) , and I will approve and copy them.
- The deadline for this is Tuesday, March 1st, so we can have plenty of time to publish the Cheat Sheet.

Things allowed in the Cheat Sheet are:

- Formulas (not solutions)
- Glossary (name in English and name in Spanish, not the definition)

Questions from last lecture

- How can we demonstrate the concept of homogeneous of degree zero
- Why, for the activities we had in class, does the solution show the marginal utility divided by the price?
- I also learned about Hicksian demand. However, I didn't understand why income is not part of its equation.
- Duda sobre las diapositivas en las que se explicaban las propiedades de continuidad, concavidad, etc., de las distintas funciones de demanda. ¿Podría explicarlo con más detalle?

5. Marshallian Demand

Indirect utility function

$V(P_1, \dots, P_n, I)$ is continuous in prices and income

$V(P_1, \dots, P_n, I)$ is **non-increasing** in prices and strictly increasing in I

$V(P_1, \dots, P_n, I)$ is convex in prices

$V(P_1, \dots, P_n, I)$ is homogeneous of degree zero in prices and income

6. Duality in Consumer Theory

Properties of the expenditure function:

- $E(P_1, P_2, \dots, P_n, U)$ is continuous in prices
- $E(P_1, P_2, \dots, P_n, U)$ is non-decreasing in prices ($\frac{\partial E}{\partial P_i} \geq 0$)
- $E(P_1, P_2, \dots, P_n, U)$ is concave in prices
- $E(P_1, P_2, \dots, P_n, U)$ is homogeneous degree one in prices
- If h_i^* is the expenditure minimizing bundle necessary to achieve utility level U at prices $(P_1, P_2, \dots, P_n)$, then:

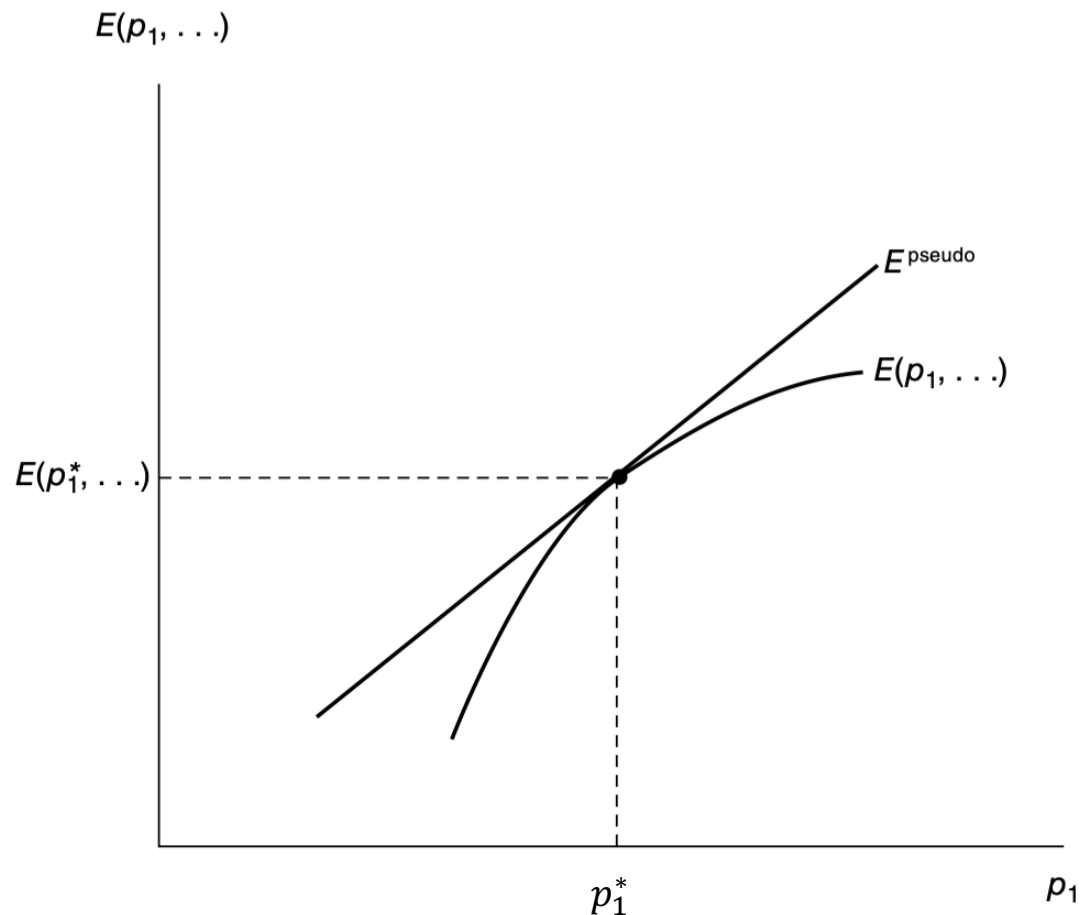
$$h_i^* = \frac{\partial E(P_1, \dots, P_n, U)}{\partial P_i}$$

for $i=1, \dots, n$ assuming the derivative exists and that $P_i > 0$

6. Duality in Consumer Theory

Properties of the expenditure function:

- $E(p_1, p_2, \dots, p_n, U)$ is concave in prices



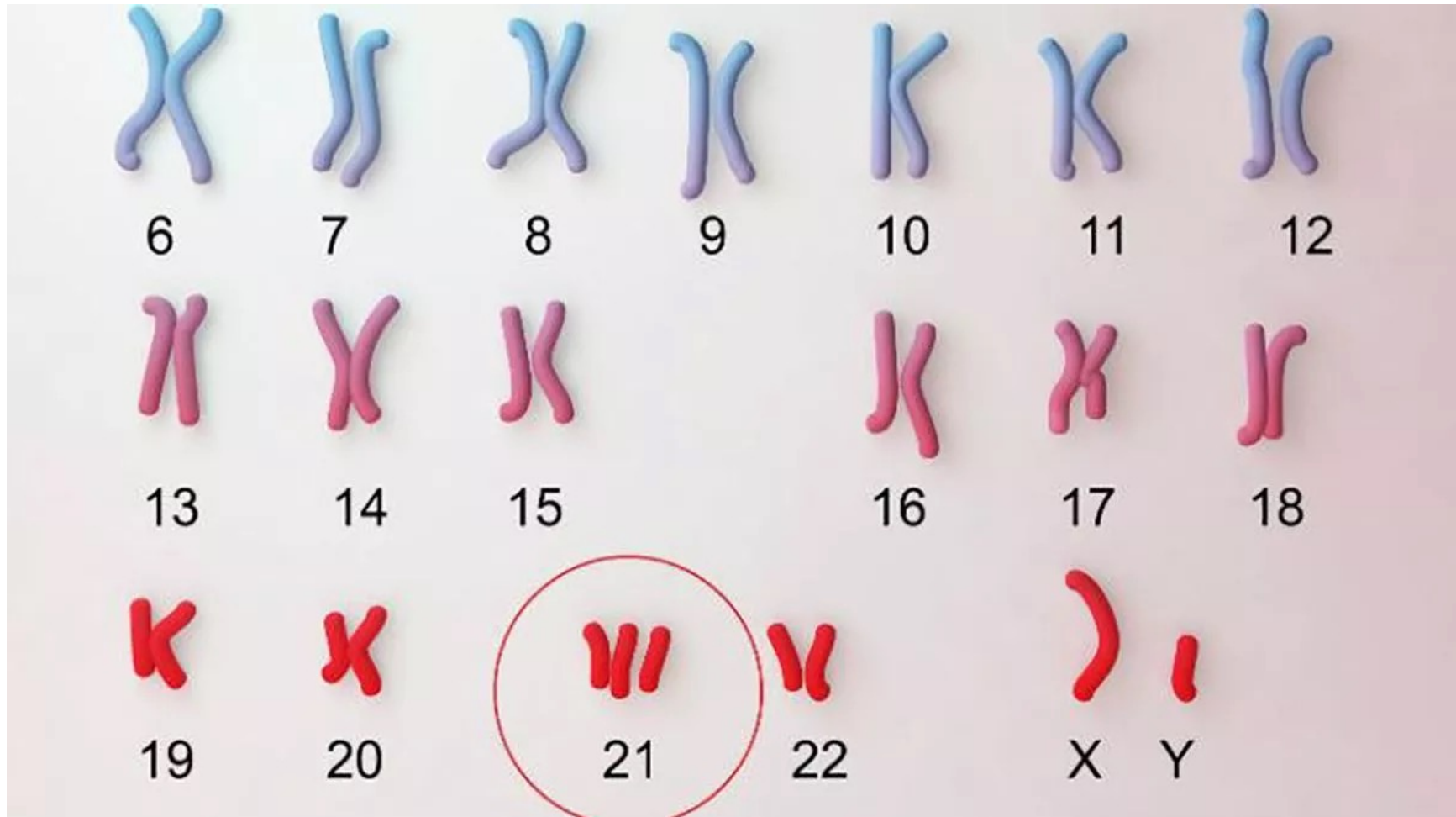


March 21st

Down
Syndrome

Awareness
day

Down syndrome: Genetic alteration



March 21st – Down Syndrome Awareness Day

The idea is to generate greater public awareness:

- Remember their inherent **dignity**.
- **Value** them as **people**.
- Value their **contributions**.
- Promote the well-being and **diversity** of communities
- Importance of **their autonomy and independence**, in particular, the freedom to make their own decisions.



March 21st

Down
Syndrome

Awareness
day

Duality in Consumer Theory

Consumer Theory

Important Identities

1. Roy's Identity:

$$x_i^* = x_i(P_1, \dots, P_n, I) \equiv - \frac{\frac{\partial V(P_1, \dots, P_n, I)}{\partial P_i}}{\frac{\partial V(P_1, \dots, P_n, I)}{\partial I}}$$

2. Shepard's lemma:

$$h_i^* = h_i(P_1, \dots, P_n, U) \equiv \frac{\partial E(P_1, \dots, P_n, U)}{\partial P_i}$$

$$3. \quad V(P_1, \dots, P_n, E(P_1, \dots, P_n, U)) \equiv U$$

Important Identities

$$4. \quad E(P_1, \dots, P_n, V(P_1, \dots, P_n, I)) \equiv I$$

$$5. \quad x_i(P_1, \dots, P_n, I) \equiv h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))$$

$$6. \quad h_i(P_1, \dots, P_n, U) \equiv x_i(P_1, \dots, P_n, E(P_1, \dots, P_n, U))$$

Example

The Cobb-Douglas function, perfect substitutes and perfect complements are special cases of the more general constant elasticity of substitution function (CES), which takes the form

$$U(x, y) = \frac{x^\delta}{\delta} + \frac{y^\delta}{\delta}$$

Where $\delta \leq 1, \delta \neq 0$ and $U(x, y) = \ln x + \ln y$ when $\delta = 0$

For an individual with a CES utility function ($\delta \neq 0$) find his/hers:

- Marshallian demand function
- Indirect utility function
- Hicksian demand function
- Expenditure function
- Demonstrate that Roy's identity and Shephard's lemma

Example: solution

a) Marshallian demand function

$$\begin{aligned} \text{Max } U(x, y) &= \frac{x^\delta}{\delta} + \frac{y^\delta}{\delta} \\ \text{Subject to: } P_x x + P_y y &= I \end{aligned}$$

Where $\delta \leq 1, \delta \neq 0$ and $U(x, y) = \ln x + \ln y$ when $\delta = 0$

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

$$\frac{\cancel{\delta} x^{\delta-1}}{\cancel{\delta}} = \frac{\cancel{\delta} y^{\delta-1}}{\cancel{\delta}}$$

$$x^{\delta-1} = \frac{P_x}{P_y} y^{\delta-1},$$

$$x = \left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} \cdot y \quad (*)$$

Example: solution

Replacing (*) in the budget constraint:

$$P_x x + P_y y = I$$

$$P_x \cdot \left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} y + P_y y = I$$

$$\left(\frac{P_x^{\delta/(\delta-1)}}{P_y^{1/(\delta-1)}} + P_y \right) y = I$$

$$y_m^* = I \frac{P_y^{1/(\delta-1)}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})}$$

Example: solution

Similarly:

$$x_m^* = I \frac{P_x^{1/(\delta-1)}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})}$$

b) Indirect utility function

Replacing $x_m^*(P_x, P_y, I)$ and $y_m^*(P_x, P_y, I)$ in the utility function:

$$V(P_x, P_y, I) = \frac{(x_m^*)^\delta}{\delta} + \frac{(y_m^*)^\delta}{\delta}$$

Example: solution

b) Indirect utility function

$$V(P_x, P_y, I) = \frac{(x_m^*)^\delta}{\delta} + \frac{(y_m^*)^\delta}{\delta}$$

$$V(P_x, P_y, I) = I^\delta \frac{P_x^{\delta/(\delta-1)}}{\delta \cdot (P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^\delta} + I^\delta \frac{P_y^{\delta/(\delta-1)}}{\delta \cdot (P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^\delta}$$

$$V(P_x, P_y, I) = \frac{I^\delta \cancel{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^1}}{\delta \cdot (P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta - \delta - 1}}$$

$$V(P_x, P_y, I) = \frac{I^\delta}{\delta} \frac{1}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}$$

Example: solution

c) Hicksian demand function

$$\text{Min} P_x x + P_y y$$

$$\text{Subject to: } U(x, y) = \frac{x^\delta}{\delta} + \frac{y^\delta}{\delta} = \bar{U}$$

Similarly to a)

$$x = \left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} \cdot y \quad (*)$$

Replacing (*) in the restriction:

$$\frac{\left[\left(\frac{P_x}{P_y}\right)^{\frac{1}{\delta-1}} \cdot y\right]^\delta}{\delta} + \frac{y^\delta}{\delta} = \bar{U}$$

Example: solution

$$\frac{\left[\left(\frac{P_x}{P_y} \right)^{\frac{1}{\delta-1}} \cdot y \right]^{\delta}}{\delta} + \frac{y^{\delta}}{\delta} = \bar{U}$$

$$\frac{\left(\frac{P_x}{P_y} \right)^{\frac{\delta}{\delta-1}} \cdot y^{\delta}}{\delta} + \frac{y^{\delta}}{\delta} = \bar{U}$$

$$y^{\delta} = \frac{\bar{U} \delta \cdot P_y^{\delta/\delta-1}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})}$$

Mental note:

$$1 + \frac{1}{\delta - 1} = \frac{\delta}{\delta - 1}$$

$$y_h^* = \frac{(\bar{U} \delta)^{1/\delta} \cdot P_y^{1/\delta-1}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}}$$

Example: solution

Similarly,

$$x_h^* = \frac{(\bar{U}\delta)^{1/\delta} \cdot P_x^{1/\delta-1}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}}$$

d) Expenditure function:

$$E(P_x, P_y, \bar{U}) = P_x \cdot x_h^* + P_y \cdot y_h^*$$

Mental note:

$$1 + \frac{1}{\delta - 1} = \frac{\delta}{\delta - 1}$$

$$= \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}} \cdot [P_x \cdot P_x^{1/\delta-1} + P_y \cdot P_y^{1/\delta-1}]$$

$$= \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{1/\delta}} \cdot [P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1}] \cdot \frac{1 - \delta}{\delta}$$

Mental note:

$$\frac{1}{\delta} - 1 = \frac{1 - \delta}{\delta}$$

Example: solution

d) Expenditure function:

$$E(P_x, P_y, \bar{U}) = \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\frac{1-\delta}{\delta}}}$$

e) Roy's identity:

$$x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}}$$

$$V(P_x, P_y, I) = \frac{I^\delta}{\delta} \frac{1}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}$$

$$x_m^* = \frac{\frac{1}{\cancel{\delta}} \frac{I^\delta}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^\delta} \cdot (+ (\cancel{\delta-1}) \cdot \frac{1}{\cancel{\delta}} / (\delta-1) \cdot P_x^{\delta/(\delta-1)-1}}{\frac{1}{\cancel{\delta}} \frac{I^{\delta-1}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}}$$

Mental note:
 $\frac{\delta}{\delta-1} - 1 = \frac{1}{\delta-1}$

Example: solution

e) Roy's identity:

$$x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}}$$

$$x_m^* = \frac{I^\delta}{I^{\delta-1}} \cdot \frac{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^{\delta-1}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})^\delta} \cdot P_x^{1/(\delta-1)}$$

$$x_m^* = I \frac{P_x^{1/(\delta-1)}}{(P_x^{\delta/(\delta-1)} + P_y^{\delta/(\delta-1)})}$$

Example: solution

e) Shephard's Lemma

$$x_h^* = \frac{\partial E(P_x, P_y, \bar{U})}{\partial P_x}$$

$$E(P_x, P_y, \bar{U}) = \frac{(\bar{U}\delta)^{1/\delta}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{\frac{1-\delta}{\delta}}}$$

Mental note:

$$\frac{\delta}{\delta-1} - 1 = \frac{1}{\delta-1}$$

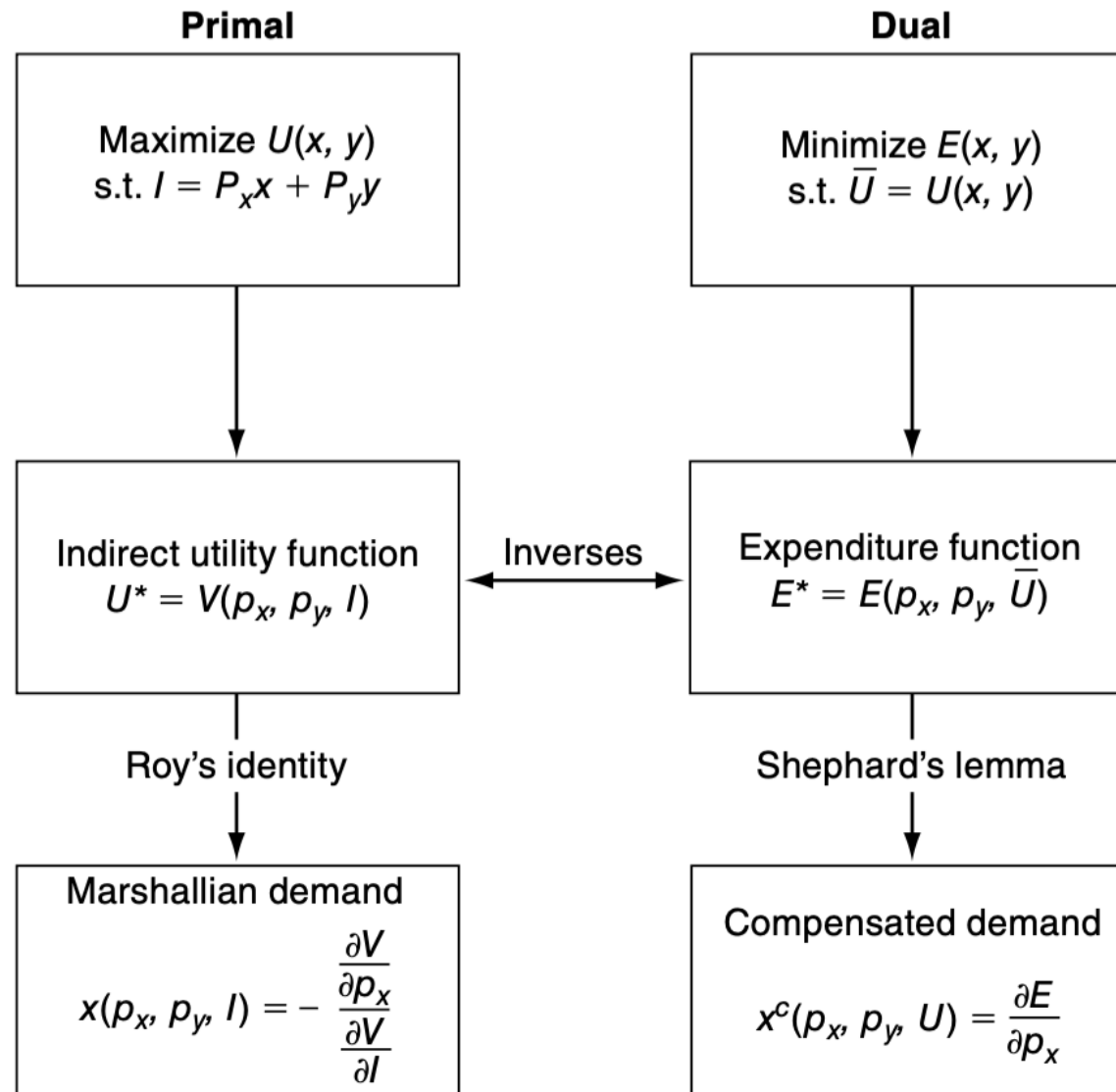
$$x_h^* = (\bar{U}\delta)^{1/\delta} \cdot \left(-\frac{1-\delta}{\delta}\right) \cdot \frac{1}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{\frac{1-\delta}{\delta}+1}} \cdot \frac{\delta}{\delta-1} \cdot P_x^{\frac{\delta}{\delta-1}-1}$$

$$x_h^* = (\bar{U}\delta)^{1/\delta} \cdot \frac{P_x^{\frac{1}{\delta-1}}}{(P_x^{\delta/\delta-1} + P_y^{\delta/\delta-1})^{\frac{1}{\delta}}}$$

Mental note:

$$\frac{1-\delta}{\delta} + 1 = \frac{1}{\delta}$$

Summary



Changes in Income

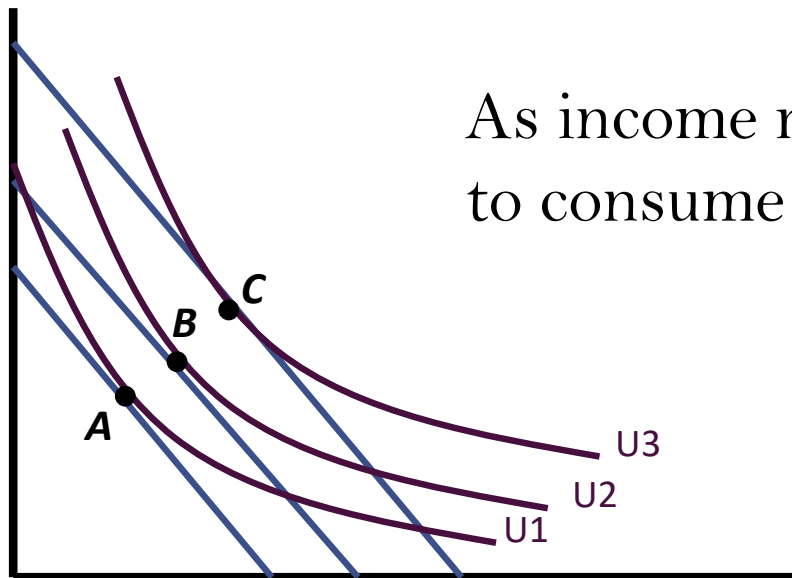
Consumer Theory

7. Changes in Income

As a person's purchasing power rises, we expect that the quantity of each good purchased will also increase. If this happens, both are **normal goods**:

$$\text{Normal goods} \rightarrow \frac{\partial x_i}{\partial I} > 0$$

Quantity of y



As income rises, the individual chooses to consume more x and y

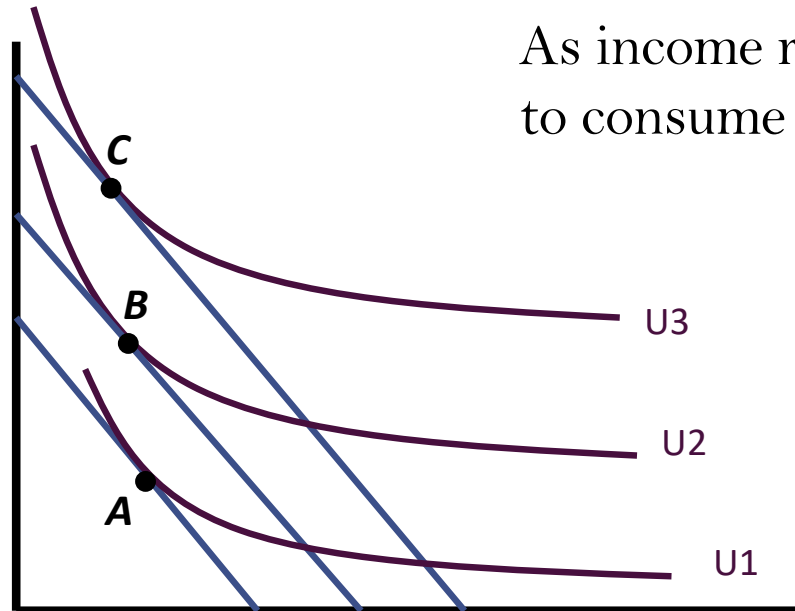
Quantity of x

7. Changes in Income

For some goods the quantity chosen may decrease as income increases and those are **inferior goods**:

Inferior goods $\rightarrow \frac{\partial x_i}{\partial I} < 0$ Examples?

Quantity of y



As income rises, the individual chooses to consume less x and more y

Note that the indifference curves do not have to be “oddly” shaped. The assumption of a diminishing *MRS* is obeyed.

Quantity of x

Activity 1

Can all goods be inferior?

1. Yes
2. No
3. It doesn't matter



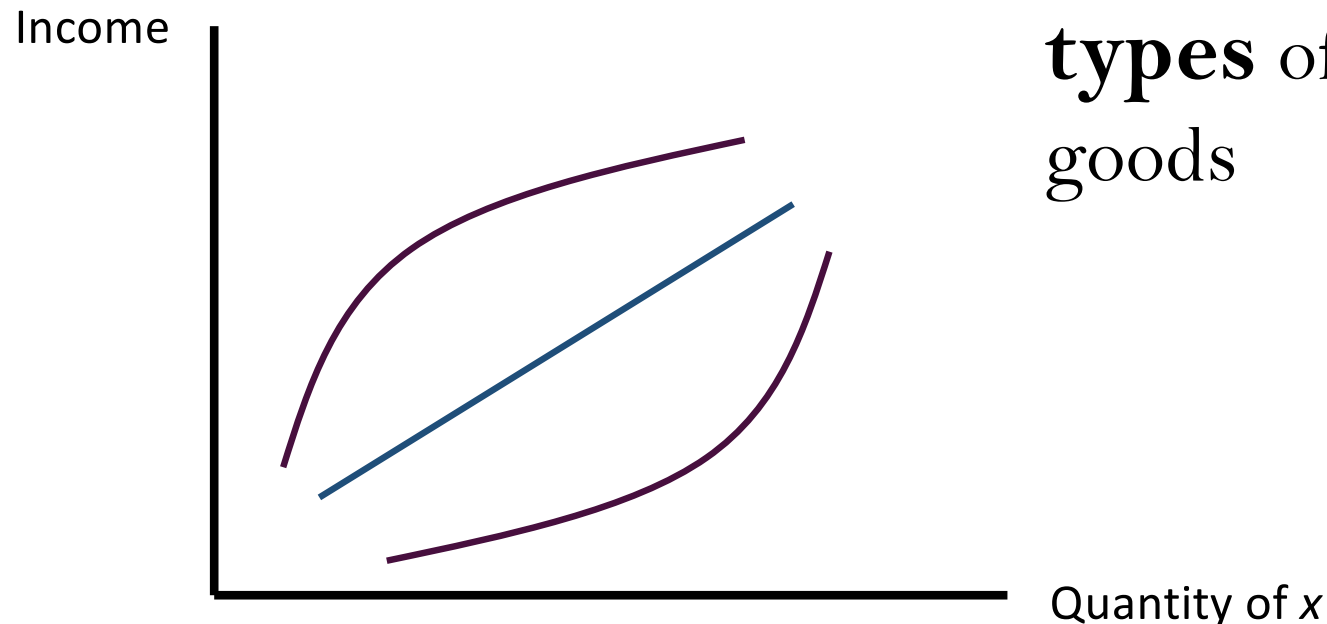
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7. Changes in Income

We can also graph **normal goods**: focusing on income and the level of consumption of that good: $\frac{\partial x_i}{\partial I} > 0$

There are **three**
types of normal
goods

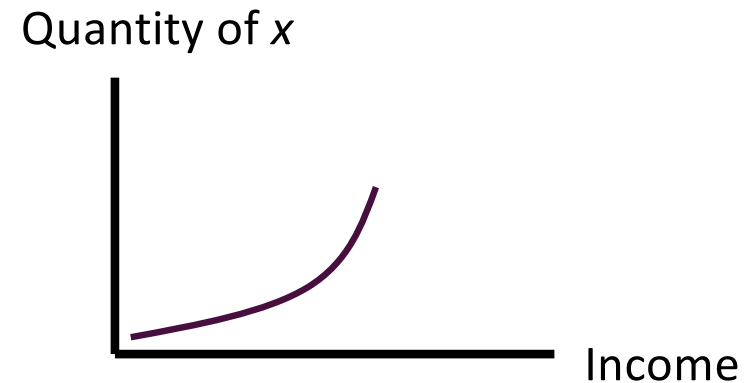


7. Changes in Income

Among the normal goods, we can also distinguish

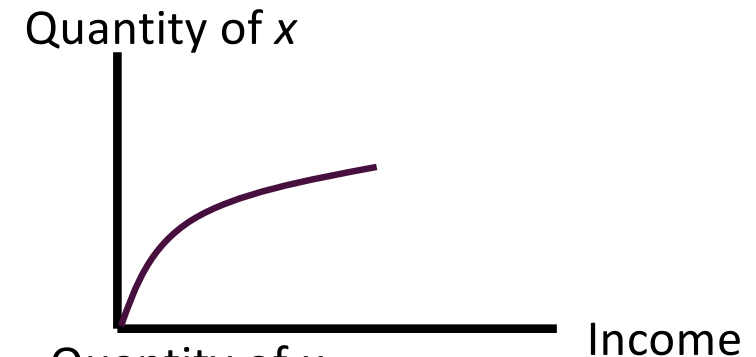
1.- **Luxury goods** →

$$\frac{\partial^2 x_i}{\partial I^2} > 0$$



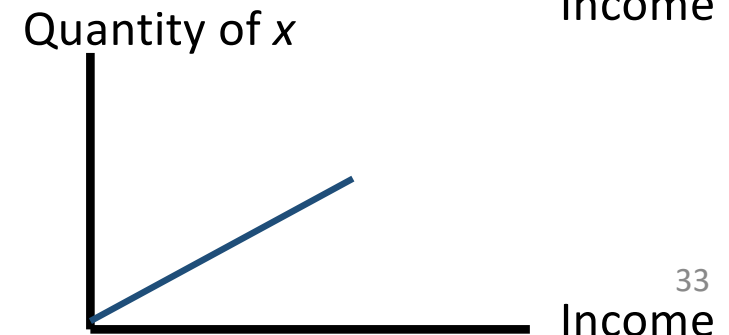
2.- **Necessary goods** →

$$\frac{\partial^2 x_i}{\partial I^2} < 0$$



3. **Homothetic goods** →

$$\frac{\partial^2 x_i}{\partial I^2} = 0$$





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Changes in good's price

Consumer Theory

8. Changes in a good's price

Since a **change in a good's price** changes the intercept of the budget constraint AND its slope, this is more complex.

Two effects come into play:

1. **Substitution Effect**
2. **Income Effect**

8. Changes in a good's price

If the **price of x decreases**

Substitution Effect: consumer should tend to buy more of that good and less from the others since they are now relatively more expensive $\rightarrow$ change on consumption of x_i associated with a change of its price at the same level of utility (i.e. same indifference curve, but now consumption patterns are allocated to equate MRS to the new price ratio)

Income Effect: since now one of the goods is cheaper, the consumer has more purchasing power $\rightarrow$ change on x_i consumption due to a change in his purchasing power (i.e., the individual cannot stay on the initial indifference curve and must move to a new one).

8. Changes in a good's price

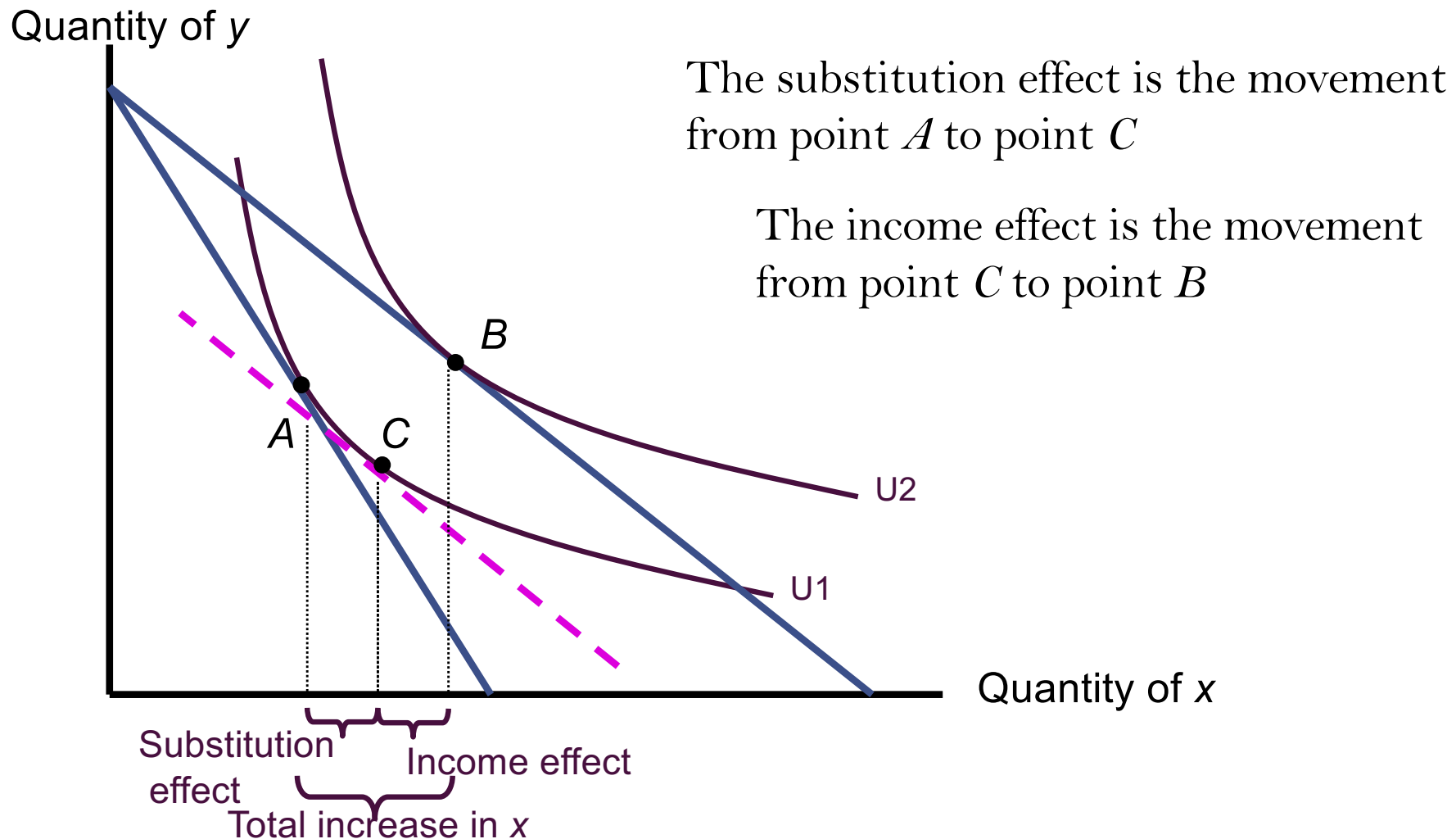
Ordinary goods: $\frac{\partial x_i}{\partial P_i} < 0$

Giffen goods: $\frac{\partial x_i}{\partial P_i} > 0$

Other real life examples of Giffen goods: [Jensen and Miller \(2008\)](#) in Hunan, China. A fall in the price of rice saved money that enabled expending more on other goods → spent more on other goods even further by buying less rice.

8. Changes in a good's price

P_x falls



Graph: Copyright ©2005 by South-Western, a division of Thomson Learning.

8. Price Changes for Normal Goods

If a good is **normal**, substitution and income effects reinforce one another

- when price falls, both effects lead to a rise in quantity demanded
- when price rises, both effects lead to a drop in quantity demanded

8. Price Changes for Inferior Goods

If a **good is inferior**, substitution and income effects move in opposite directions

The combined effect is indeterminate

- when price rises, the substitution effect leads to a drop in quantity demanded, but the income effect is opposite
- when price falls, the substitution effect leads to a rise in quantity demanded, but the income effect is opposite

8. Giffen's Paradox

If the income effect of a price change is strong enough, there could be a positive relationship between price and quantity demanded

- an increase in price leads to a drop in real income
- since the good is inferior, a drop in income causes quantity demanded to rise

8. Changes in a Good's Price

Substitution Effect:

- If $P_x \uparrow \rightarrow \downarrow x$
- If $P_x \downarrow \rightarrow \uparrow x$

Income Effect: Depends on the type of good

- If $P_x \uparrow$ ($\downarrow$ purchasing power)
 - If it is a normal good $\rightarrow \downarrow x$
 - If it is an inferior good $\rightarrow \uparrow x$
- If $P_x \downarrow$ ($\uparrow$ purchasing power)
 - If it is a normal good $\rightarrow \uparrow x$
 - If it is an inferior good $\rightarrow \downarrow x$
- Giffen goods are a subset of inferior goods, where the Income Effect is stronger than substitution effect.

Activity 2

If the price of x decreases:

1. What happens to the consumption of x if it is a normal good?
 - a. Rises
 - b. Falls
 - c. Nothing
 - d. It depends
2. And what happens to the consumption of x if it is an inferior good?
 - a. Rises
 - b. Falls
 - c. Nothing
 - d. It depends



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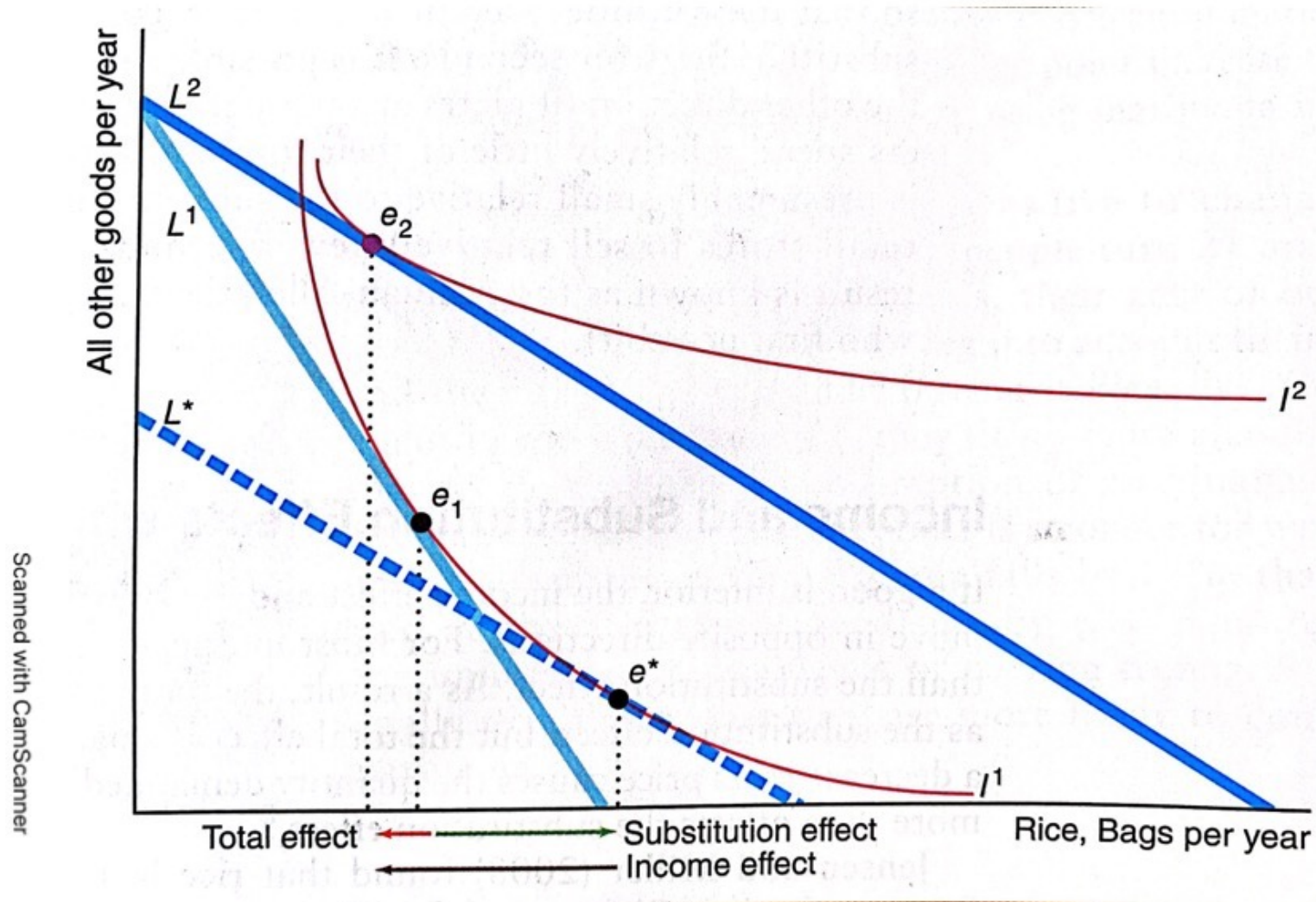
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Activity 3

Analyse graphically what happens with x consumption if there is a decrease of P_x and x is Giffen good.



Giffen example (inspired in rice)



Slutsky Equation

Consumer Theory

9. Slutsky equation

The **Slutsky Equation** is a mathematical development of response to price changes

We wish to calculate $\frac{\partial x_i}{\partial P_j}$

Slutsky equation:

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_j} = \frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_j} - \frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_j(P_1, \dots, P_n, I)$$

9. Slutsky equation

Proof:

Let x^* maximize utility at (P^*, m^*) and let $U^* = U(x^*)$.
From identity (6) from Lecture 5

$$h_i(P_1, \dots, P_n, U^*) \equiv x_i(P_1, \dots, P_n, E(P_1, \dots, P_n, U^*))$$

If we differentiate this with respect to P_j and evaluate the derivative at P^*

$$\frac{\partial h_i(P_1^*, \dots, P_n^*, U)}{\partial P_j} = \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial P_j} + \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial I} \boxed{\frac{\partial E(P_1^*, \dots, P_n^*, U^*)}{\partial P_j}}$$

Consumer Theory

Since
$$\frac{\partial E(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} = x_j$$

Slutsky equation:

$$\frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial P_j} = \frac{\partial h_i(P_1^*, \dots, P_n^*, V(P_1^*, \dots, P_n^*, I^*))}{\partial P_j} - \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial I} x_j(P_1^*, \dots, P_n^*, I^*)$$

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.